

Class XI – Mathematics

Model Answer Sheet (Full Marks)

Section A – MCQs

1. (b) 8
2. (b) Contains all elements under consideration
3. (a) $\frac{4}{5}$
4. (c) Many–one relation
5. (b) $\frac{1}{2}$
6. (c) 60
7. (b) $A \cap B = \emptyset$
8. (c) Transitive

Section B – Very Short Answer

9. A set is a well-defined collection of objects. Example: $A = \{1, 2, 3\}$.
10. Domain = $\mathbb{R}$, Range = $\{x : x \geq 0\}$.
11. $\sin 45^\circ \cos 45^\circ = (\frac{1}{\sqrt{2}})(\frac{1}{\sqrt{2}}) = \frac{1}{2}$.
12. $nP_4 = 5 \times 4 = 20$.
13. Number of elements in power set = $2^n = 16$.
14. (i) One–one: $f(x) = x + 1$
(ii) Many–one: $f(x) = x^2$

Section C – Short Answer

15.
$$\begin{aligned} \text{LHS} &= \left(\frac{\sin\theta}{1 + \cos\theta} \right) + \left(\frac{1 + \cos\theta}{\sin\theta} \right) \\ &= \frac{(\sin^2\theta + (1 + \cos\theta)^2)}{\sin\theta(1 + \cos\theta)} \\ &= \frac{(\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta)}{\sin\theta(1 + \cos\theta)} \\ &= \frac{(2 + 2\cos\theta)}{\sin\theta(1 + \cos\theta)} \\ &= \frac{2}{\sin\theta} = 2 \operatorname{cosec}\theta = \text{RHS. Hence proved.} \end{aligned}$$

16. (i) $A \cup B = \{1, 2, 3, 4, 5, 6\}$

(ii) $A \cap B = \{3, 4\}$.

17. Number of 3-digit numbers = ${}^5P_3 = 5 \times 4 \times 3 = 60$.

18. The relation is reflexive since $(a,a) \in R$ for all $a \in A$. It is symmetric and transitive. Hence, it is an equivalence relation.

Section D – Long Answer

19. $\text{LHS} = (1 + \tan^2\theta)(1 + \sin^2\theta)$
 $= \sec^2\theta(1 + \sin^2\theta)$
 $= \text{RHS. Hence proved.}$

20. Letters in MATHEMATICS = 11

$M = 2, A = 2, T = 2$

Number of arrangements = $11! / (2!2!2!) = 4,989,600$.